

COURSE TITLE : FURNITURE TECHNOLOGY
COURSE CODE : 6122
COURSE CATEGORY : A
PERIODS PER WEEK : 5
SEMESTER : 6
PERIODS PER SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
I	History of Furniture	2
	Hand Tools	3
	Wood Joints	5
	Stationary Machines	7
	Test	1
II	Portable machines	6
	Jigs and Fixtures	6
	Furniture Hardware	6
	Test	1
III	CNC Machines	6
	Project Planning	6
	Preparation For Finishing	6
	Test	1
IV	Furniture Finishing	18
	Test	1
Total		75

Course Outcome :

Module	G.O	Student will be able to:
1	1	Know Furniture Technology
	2	Understand Machines Used For Mass Production of Furniture
2	1	Understand Portable Machines
	2	Understand The Classification of Joints, Jigs And Fixtures
3	1	Understand CNC Machines
	2	Understand Project Planning
	3	Understand Preparation of Furniture For Finishing
	4	Understand Sanding

4	1	Understand Finishing of Furniture
	2	Understand Polishing

Specific outcome

MODULE-I

1.1.0 Know Furniture Technology

- 1.1.1 Discuss the history and development of furniture
- 1.1.2 Discuss the raw materials used for furniture solid, wood, plywood veneer block board, particle board and glue
- 1.1.3 Discuss the selection and characteristics of hand tools used in furniture

1.2.0 Understand Machines Used For Mass Production of Furniture

- 1.2.1 Discuss the machinery for mass production of furniture
- 1.2.2 Discuss the circular saw, jig saw narrow, band saw thickness planer, jointer spindle molder router tenoning machine chain and chisel mortiser doweling machine – important parts and their functions

MODULE-II

2.1.0 Understand Portable Machines

- 2.1.1 Discuss portable machines used in furniture
- 2.1.2 Describe Portable drilling machine, portable jigsaw, portable circular and sanders
- 2.1.3 Understand Furniture Design and Production
- 2.1.4 Discuss the furniture designing
- 2.1.5 Describe the fundamentals of design
- 2.1.6 Discuss various types of design tools

2.2.0 Understand the Classification of Joints, Jigs and Fixtures

- 2.2.1 Classify joints
- 2.2.2 Explain lengthening, widening, framing and box joints
- 2.2.3 Discuss jigs and fixtures
- 2.2.4 Know Furniture Hardware
- 2.2.5 Discuss hinges, screws, nails, locks and accessories

MODULE-III

3.1.0 Understand CNC Machines

- 3.1.1 Introduction to CNC
- 3.1.2 Components of NC systems
- 3.1.3 Part programming machine, control unit, machine tools, advantages of CNC
- 3.1.4 CNC Technology
- 3.1.5 CNC machine fundamentals

3.1.6 Simple Problems

3.2.0 Understand Project Planning

- 3.2.1 Discuss preparation of bill of materials
- 3.2.2 Estimate the cost of furniture
- 3.2.3 Prepare plant and machine layout of furniture factory
- 3.2.4 Demonstrate wood carving
- 3.2.5 Discuss carving tools
- 3.2.6 Prepare furniture drawing

3.3.0 Understand Preparation of Furniture For Finishing

- 3.3.1 Discuss preparation for finishing such as removing excess glue
- 3.3.2 Discuss preparing wood surfaces with rot Iron and cloth Burn in method filler materials, wood pegs or plugs
- 3.3.3 Discuss bleaching and kinds of bleaches
- 3.3.4 Describe safety in bleaching and application of bleaches

3.4.0 Understand Sanding

- 3.4.1 Describe – coated abrasives
- 3.4.2 Describe flint garnet aluminium oxide silicon carbide and grading of abrasives
- 3.4.3 Discuss the use of sand papers
- 3.4.4 Discuss the use of stationery and portable sanders
- 3.4.5 Use portable sander – finishing portable belt sander & rotary active sander
- 3.4.6 Use stationary sanders – belt sander, disc, sander, spindle sander and hand stroke belt sander
- 3.4.7 Discuss hand sanding – Important steps to be followed in sanding

MODULE-IV

4.1.0 Understand Finishing of Furniture

- 4.1.1 Describe sap staining, ingredients advantages and disadvantages apply of wash coat
- 4.1.2 Explain oil stain – ingredient advantages and disadvantages apply of NGR stain
- 4.1.3 Explain non grain raising (NGR) stain – advantages & disadvantages
- 4.1.4 Explain non – grain – raising (NGR) stain – advantages & disadvantages – apply of NGR stain
- 4.1.5 Describe wash coat – its purpose- apply of wash coat
- 4.1.6 Describe filling
- 4.1.7 Discuss kinds of fillers – paste & liquid – applying paste & liquid fillers
- 4.1.8 Describe sealing and applying a sealer
- 4.1.9 Describe finishing
- 4.1.10 Describe oil finishes - linseed oil finish – applying oil finish
- 4.1.11 Describe shellac – uses and types of shellac

4.2.0 Understand Polishing

- 4.2.1 Describe French polishing
- 4.2.2 Describe cellulose lacquer
- 4.2.3 Explain various ingredients and kinds of lacquer
- 4.2.4 Describe advantages and disadvantages of lacquer finish
- 4.2.5 Discuss the usage of lacquer thinners & lacquer finishing schedule
- 4.2.6 Describe the process of staining and filling before applying lacquer
- 4.2.7 Describe the usage of sealers for lacquer
- 4.2.8 Describe the process of applying spray lacquer finish
- 4.2.9 Discuss the process of applying a clear brushing lacquer finish
- 4.2.10 Discuss the types of finishes like polymethanes, polyester, epoxy
- 4.2.11 State the purpose of rubbing & polishing
- 4.2.12 Discuss the process of rubbing, polishing and cleaning
- 4.2.13 Discuss the materials used in rubbing & finishing

CONTENT DETAILS

MODULE-I

General consideration of furniture technology. History & development of furniture – Raw materials – solid wood, plywood, veneer, block board, glue – preparation and characteristics – hand tools used in furniture making- Machines used for mass production of furniture (Brief ideas only) Circular saw, jig saw narrow band saw, thickness planner, jointer, spindle moulder, router, tenoning machine chain and chisel mortiser, Doweling machine

MODULE-II

Portable machines – portable drilling machine, portable jig saw, portable circular saw and sanders Saw doctoring – teeth geometry – tensioning, joining methods of saw – saw setting Furniture designing & production Fundamentals of design – design tools, planning a project, classification of joints – lengthening – widening, framing and box joints Jigs and fixtures Furniture hardware – hinges, screws, nails, locks, and accessories Preparation of bill of materials – estimating and costing of furniture Plant and materials layout; of furniture factory carving – tools used for carving Assignment – preparation of drawing of furniture

MODULE-III

Portable machines – portable drilling machine, portable jig saw, portable circular saw and sanders Saw doctoring – teeth geometry – tensioning, joining methods of saw – saw setting Furniture designing & production Fundamentals of design – design tools, planning a project, classification of joints – lengthening – widening, framing and box joints Jigs and fixtures Furniture hardware – hinges, screws, nails, locks, and accessories Preparation of bill of materials – estimating and costing of furniture Plant and materials layout; of furniture factory carving – tools used for carving Assignment – preparation of drawing of furniture

MODULE-III

Introduction to CNC- components of NC system- part programming, machine control unit, machine tools, advantages of CNC, CNC technology, CNC machine fundamentals, Simple problems. Furniture finishing – preparation for finishing – removing excess glue – repairing wood surfaces with hot iron and cloth – burn in method, filler materials – wood pegs or plugs, bleaching - kind of bleaches – safety –

applications Sanding –coated abrasives –flint – garnet aluminium oxide, silicon carbide, grading of abrasives. Sand paper grit size – closed and open coat abrasives. Stationery and portable sander. – Floor type belt sander- Disc sander – spindle sander, hand stroke belt sander. Portable sanders – finishing sanders, portable belt sanders, rotary action sanders important steps to be followed in Finishing-Project Planning-Estimating and Costing-Calculations

MODULE-IV

Staining and filling-sap staining – ingredients in stains-purpose of staining kinds of stain- water, oil and NGR stains- applying oil stains advantages and disadvantages (of above stains) and their ingredients. Wash coat – purpose – applying wash coat. Filling – kinds of fillers – paste and liquid fillers. Applying paste and liquid fillers. Sealing – applying a sealer Protective coating – oil finishers – linseed oil finisher – applying oil finish. Shellac – uses – types of Shellac- French polishing – Cellulose lacquers – ingredients. Advantages And disadvantages. Kinds of lacquers. – Lacquer thinners – lacquer finishing schedule – staining and filling before applying lacquer – sealers for lacquer – Applying a spray lacquer finish spraying a project- Applying a clear brushing lacquer finish. An introduction to plastic lacquers, polyurethane, polyester and epoxy finisher. Rubbing, polishing and cleaning – purpose of rubbing and polishing – materials used for rubbing and polishing.

REFERENCE

1. John A Walton, Wood works in theory and practice –, Australian publishing co.
2. Carpentry a complete guide – McGraw Hill, New Delhi.
3. Eric Stephensor and Dave plank, Rochdati ,Circular Saw –, Thom Robinsons and Son Ltd. England
4. Robert Schalff, Complete Book of Wood finishing –
5. Ernest Scott. Working in wood –
6. Joseph De Chiala, Time Saver standards for interior Design –McGraw Hill, New York.
7. P.C Sharma ,Production Technology
8. Ahamed A. Kasu ,An introduction to Art and Craft Science - (Ashish Book Cen. Mumbai)